

LING 510: Week 1 review

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Office hours: W 11–1PM, N421H | [Zoom](#)



Plan for the day: Review concepts, practice, practice, your questions

1 Compositionality

- The Semantics that you will be learning in this class is known as *compositional semantics*.
- The founding assumption of compositionality is that the meaning of a sentence is determined by the meaning of its parts...
 - (1) **Parts matter for meaning**
 - a. Eugenides stole Attolia's earrings.
(This sentence describes a situation that occurred some time before now in which a person called Eugenides engaged in the act of stealing earrings that belonged to a person called Attolia.)
 - b. Eugenides stole Attolia's crown.
(This sentence describes a situation that occurred some time before now in which a person called Eugenides engaged in the act of stealing a crown that belonged to a person called Attolia.)
 - ...and the order/the way in which those parts combine.
 - (2) **Order of the parts matters for meaning**
 - a. Attolia stole Eugenides' earrings.
This sentence describes a situation that occurred some time before now in which a person called Attolia was the agent of the act of stealing earrings that belonged to a person called Eugenides.
 - b. Eugenides stole Attolia's earrings.
This sentence describes a situation that occurred some time before now in which a person called Eugenides was the agent of the act of stealing earrings that belonged to a person called Attolia.
- Sentences will have parts but sometimes some parts will have parts of their own.
- So we need two things to determine the meaning of a sentence: 1. Lexicon 2. Syntax.
- Assuming you don't know who I am talking about in (1) and (2), you still understood the meaning of the sentences. The paraphrases are how you would tell me the meanings. but what the heck does it mean to understand the meaning of a sentence?

2 Truth conditions

- When you hear a sentence, you know what to look for out in the world to find the situation that matches the situation described by the sentence.
- This is a pretty trivial-looking but actually a weapons-grade insight.
- David Lewis said *meaning is as meaning does*. (Lewis 1976)
- “In order to say what a meaning is, we may first ask what a meaning does, and then find something that does that”
- To agree/disagree with me about (1), you would need to know what to look for in the world and find the situation does or does not match the situation described by the sentence.
- When you know the meaning, you know what sort of situation would verify the sentence i.e., when sentences will be true (and when it will be false).
- Note that knowledge of meaning does not require you to know whether a sentence is true or false. It requires you to know **when the sentence would be true** (or false).
- A theory of meaning would be like a machinery that would take a sentence and its parts and the rules of combining those parts and derive the right truth conditions of a sentence.

To become comfortable with this notion of truth conditions, we will do some exercises. Table 1 describes a situation *s* involving 5 teletubbies called Tinky-winky, Dipsy, Laa-laa, Po, and 4 tiddlytubbies Mi-Mi, Da-Da, Baa, Ru-Ru, and Ping.

<i>Teletubby</i>	Situation <i>s</i>	
	<i>Occupation</i>	<i>Preference</i>
Tinky-winky	leader	red bags, playtime
Dipsy	student	dancing, playtime
Laa-laa	dancer, student	singing and dancing, playtime
Po	student	driving, playtime
Mi-Mi	baby	sleeping, playtime
Da-Da	baby	sleeping
Baa	baby	sleeping and singing
Ru-Ru	baby	singing
Ping	baby	sleeping

Table 1: teletubbies

- (3) a. At least two teletubbies are students.
 b. A student who likes playtime likes something else too.
 c. few teletubbies like sleeping.
- (4) a. If a teletubby likes sleeping then it is a baby.
 b. If a teletubby likes playtime then it is a baby.
 c. Every teletubby that likes singing also likes dancing.
 d. Many teletubbies like driving.

Question 1. Are the sentences in (3) and (4) true in s . Explain your reasoning.

For instance, the sentence *Teletubbies that like singing are either a baby or a dancer* is true in s because Ru-Ru and Baa like singing and they are babies, Laa-Laa also likes singing and she is a dancer. These are the only teletubbies in s that like singing.

If a sentence is false in s , describe a situation (using a table) which makes it true. Specify which situation makes which sentence true.

Note: The description should contain just enough information to make it true.

3 Entailments

Truth conditions allow us to neatly capture logical relationships between sentences.

- (5) a. Laa-Laa likes singing and dancing.
 b. Laa-Laa likes singing.

Observation: When (5-a) is true, (5-b) automatically has to be true.

Entailment: For sentences S_1 and S_2 , *whenever* S_1 is true, S_2 is also true, we say that S_1 entails S_2 .

- This means that the truth conditions of S_1 are also always the truth conditions of S_2 !

(6-a) entails (6-b) (check!). What happens to truth value of (6-a) if (6-b) were to be false?

- (6) a. At least two teletubbies are students.
 b. A teletubby is a student.

(Hint: (6-b) is false when its negation is true.)

Note the emphasis on ‘whenever’. That means, if you find S_2 to be only sometimes (or never) true while keeping S_1 true, there is no entailment between S_1 and S_2 .

Corollary: Suppose S_1 entails S_2 , then whenever S_2 is not true then S_1 is also not true.

- Putting the definition and the corollary together we get a diagnostic to determine entailment relationships.
- we ask what happens when S_1 is true and S_2 is not true?

Contradiction test for entailment

S_1 entails S_2 , if and only if the sentence ' S_1 and/but it is not the case that S_2 ' is a contradiction.

Demonstration:

(7) #*Laa-Laa likes singing and dancing but Laa-Laa doesn't like singing.*

There is no situation which would make (7) true. Therefore, it is a contradiction.¹

Question: Apply the contradiction test to determine whether (6-b) entails (6-a)? If not, describe a situation in which both aren't true simultaneously. What about the following pairs?

- (8) a. A teletubby is a student.
b. Some teletubby is a student.
- (9) a. John was not aware that he passed the exam.
b. John passed the exam.
- (10) a. Not every student likes Semantics.
b. No student likes Semantics.
- (11) a. Fewer than 3 babies like dancing.
b. At most 2 babies like dancing.

You will have noticed that sometimes two sentences entail each other i.e., they have identical truth conditions – Every situation that makes S_1 true, makes S_2 true and vice versa.

Also note that if two sentences mean (intuitively) the same thing then they entail each other. Verify! (same meaning → same truth conditions → entailment)

- (12) a. John is a bachelor.
b. John is an unmarried man.

Logical equivalence

If S_1 entails S_2 and S_2 entails S_1 then S_1 and S_2 are said to be logically equivalent.

Corollary: If there is a situation s in which S_1 is true but S_2 is false (or simply that they have different truth values) then they are not logically equivalent.

Question: Which sentences from (8)–(11) are equivalent?

Now consider the entailment patterns for this data. How does it change?

- | | |
|---|---|
| (13) a. John read a Russian book.
b. John read a book. | (16) a. Possibly, John read a Russian book.
b. John read a book. |
| (14) a. Allegedly, John read a Russian book.
b. John read a book. | (17) a. John rarely reads Russian books.
b. John rarely reads books. |
| (15) a. John often reads Russian books.
b. John often reads books. | (18) a. John read every Russian book.
b. John read every book. |

¹Contradictions are always false.

4 Ambiguity

Sometimes a sentence is pronounced the same way but it can have distinct (literal) meanings.

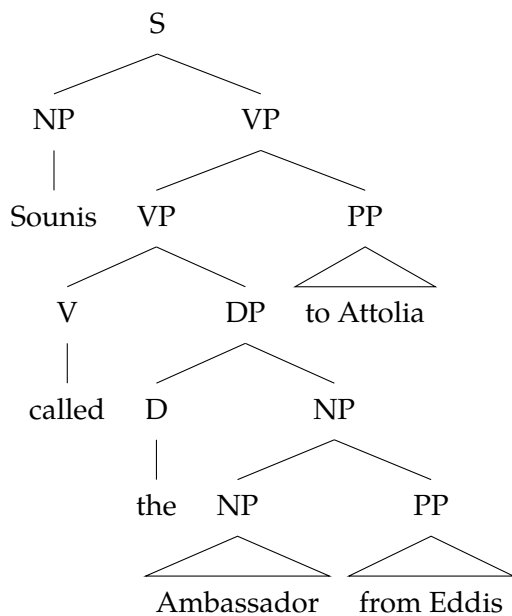
(19) Sounis called the Ambassador from Eddis to Attolia.

When a sentence is ambiguous the two parses are said to correspond to two *readings*. (19) means:

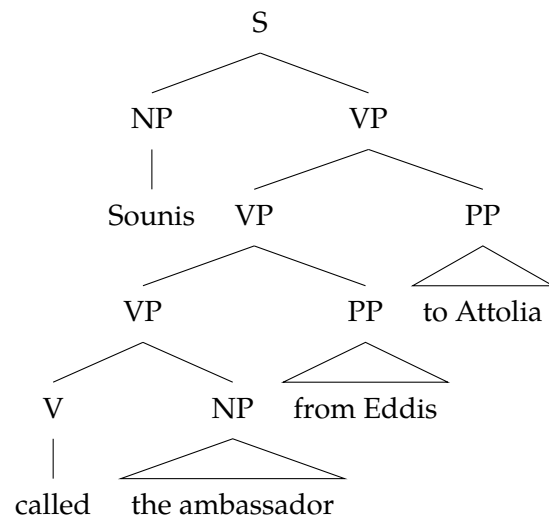
- (20) a. Sounis called the Ambassador who is from Eddis to Attolia.
b. Sounis called the Ambassador to Attolia when Sounis was in Eddis.

Each meaning corresponds to a different structure i.e., the ambiguity in the meaning is due to the ambiguity in the structure.

(a) *The Ambassador who is from Eddis*



(b) *Sounis was in Eddis when he called*



Question: Can you tell which reading entails *Sounis was in Eddis*? Do either of the readings entail *Sounis was in Attolia*?

Sometimes the source ambiguity is not structural but rather because there two different ways to interpret a word. This is **lexical ambiguity**.

- (21) Comic books are not dirty.
↪ *Comic books are not unclean.*
↪ *Comic books are not morally improper and obscene.*

Notice that we began by saying that meaning of parts and the way they combine matters for meaning. The fact that both are independently sources of ambiguities further proves this! Neat!

References

Lewis, David. 1976. General semantics. In *Montague grammar*, 1–50. Elsevier.